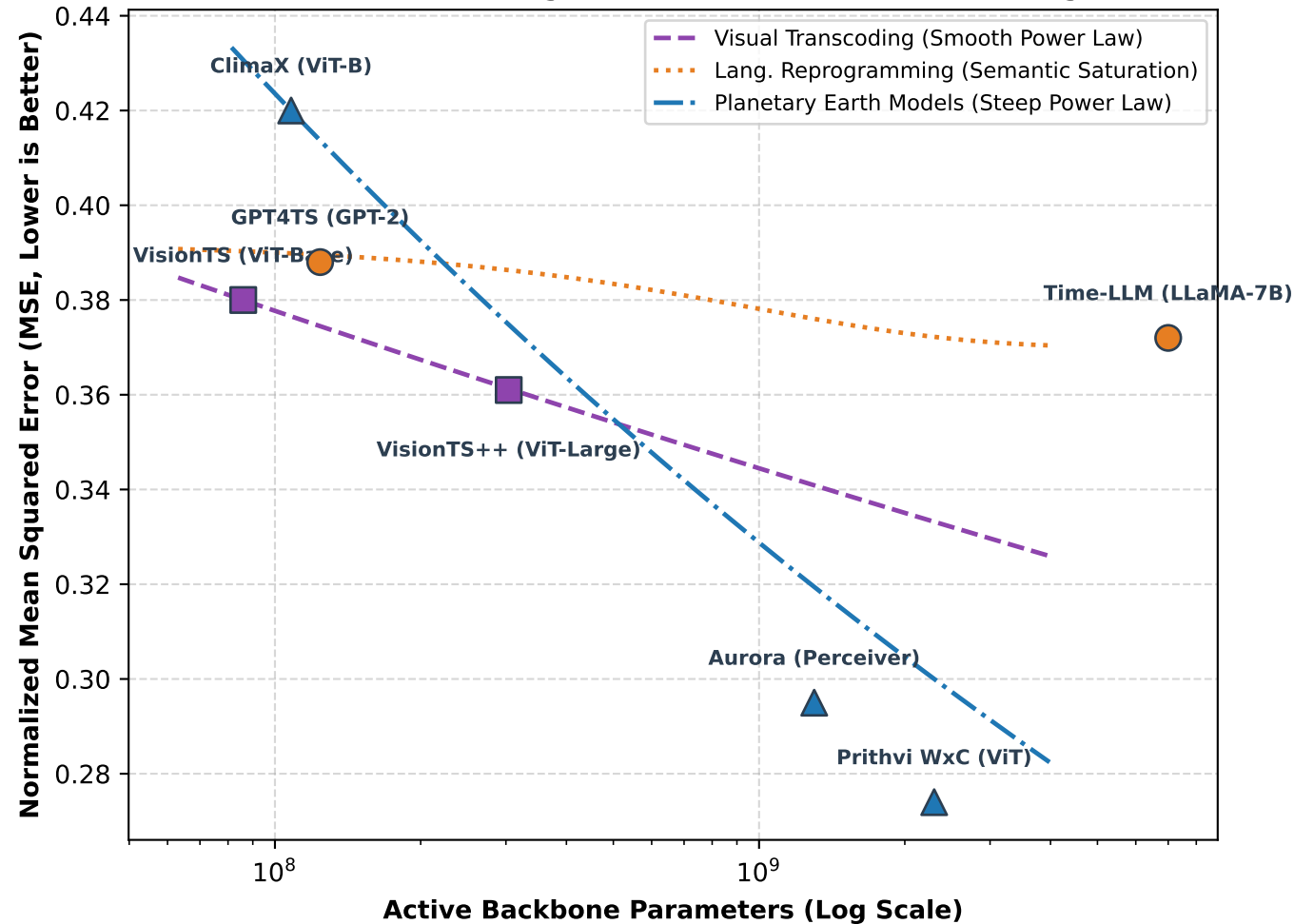


(a) Parameter Scaling Laws Across Multimodal Paradigms



(b) Pre-training Token Scaling & Semantic Gain Advantage

